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Working Group IX - Characteristics of Cosmological Galaxies

Welcome to team IX! Please review the following action items and send us any comments or questions you may have.

Leadership

Cameron Hummels and Javiera Guedes share the leadership of this team.

Participants

Oscar Agertz, Daniel Ceverino, María Emilia De Rossi, Fabrice Durier, Javiera Guedes, Cameron Hummels, Jason Jaacks, Dusan Keres, Andrey Kravtsov, Sam Leitner, Lucio Mayer, Daisuke Nagai, Ken Nagamine, Brian O'Shea, Joel Primack, Justin Read, Brant Robertson, Emilio Romano-Díaz, Rok Roskar, Wolfram Schmidt, Sijing Shen, Britton Smith, Robert Thompson, Matteo Tomassetti, Robert Feldmann.

Science Topics

Our working group will focus on the following science questions:

- What are the structural properties of the simulated galaxies at $z=0$, and how do these properties compare to observations? This involves measuring rotation curves, B/D ratios (from photometry and kinematics), surface brightness profiles, star formation histories, X-ray emission and/or dispersion measure from hot gas content, HI rotation curves, colors, bar strength ($m=2$ Fourier mode), etc.
- What is the origin and evolution of the principal galactic structures? How does the disk, bulge, stellar halo, and thick disk form and evolve?
- What are the hot / cold gas budgets and main accretion modes?
- What is the role of stellar migration in the evolution of galactic disks?
- Do we see metallicity gradients in the disk / thick disk? How did these gradients come about?

Let us know of other topics you would like to investigate!

Code Representation

As far as we can tell from the email list, we have representatives from:

Arepo?: Keres?

Art: Ceverino

Chicago-Art: Leitner

Enzo: Hummels

Gasoline: Guedes, Shen, Roskar

Gadget: Keres, Nagamine, Thompson, De Rossi

Gadget + SPHS: Read

Ramses: Agertz

Nyx: Schmidt

Please sign up / correct this list.

Analysis Tools

Some of the tools we will be using on this project are:

yt: Hummels

BC03: Hummels
Galfit: Guedes
SUNRISE: Guedes

Please sign up if you are comfortable running these tools, and/or add more.